Logarithm Plots: NC COVID Hospitalization Report Guide

1. Executive Overview & Project Objective

- **Data Set:** 7-day running average of daily COVID-19 hospitalizations in North Carolina starting April 14, 2020 ($t = 0$).
- **Goal:** Model non-linear, multi-phase epidemiological growth and containment using **piecewise semi-log regression**.
- **Core Transformation:** An exponential curve $y = Ae^{kt}$ becomes a linear function when plotted on a logarithmic scale:

$$\ln(y) = \ln(Ae^{kt}) = \ln(A) + kt = mt + b, \quad \text{where } m = k \text{ and } b = \ln(A).$$

2. Mathematical Preliminaries: Piecewise Linear Functions

A function is **piecewise linear** if its domain can be partitioned into intervals such that the function is strictly linear on each piece:

$$f(x) = \begin{cases} x + 4 & \text{when } 0 < x < 4 \\ 2x + 1 & \text{when } 4 \leq x < 8 \end{cases} \quad \textbf{(Piecewise Linear)}$$

$$g(x) = \begin{cases} x + 4 & \text{when } 0 < x < 4 \\ x^2 & \text{when } 4 \leq x < 8 \end{cases} \quad \textbf{(Not Piecewise Linear)}$$

Key Insight for Semi-Log Modeling: A curve $y(t)$ whose logarithm $\ln(y(t))$ is piecewise linear corresponds to a **piecewise exponential model** in real terms, where each regime has a constant relative growth/decay rate k.

3. Google Sheets Setup & Data Structure

- **Column A (Date):** Calendar date starting 4/14/2020.
- **Column B (Days since 4/14/2020, t):** Indexed time $t = 0, 1, 2, \ldots$
- **Column C (Hospitalizations, y):** Raw 7-day moving average (e.g., $t = 0 \implies 375.86$, $t = 1 \implies 382.29$).
- **Column D ($\ln$(hosp)):** Natural log of Column C via formula `=LN(C2)`.

4. Piecewise Log Regressions (NC COVID-19 Data)

Phase 1: Initial Spring Wave (4/14/2020 – 5/5/2020, $t \in [0, 21]$)

- Google Sheets Trendline Fit: $\ln(y) = 0.0157\,t + 5.93$
- **Recovery of Exponential Model:**

$$y(t) = e^{\ln(y)} = e^{0.0157\,t + 5.93} = e^{5.93} \cdot e^{0.0157\,t} \approx 376.15 \cdot e^{0.0157\,t}$$

- **Epidemiological Interpretation:**
  - Base initial hospitalization count at $t = 0$: $A \approx 376$ patients.
  - Daily continuous growth rate: $k = 0.0157$ (+1.57% daily growth).
  - Doubling time: $T_{\text{double}} = \frac{\ln(2)}{k} = \frac{0.6931}{0.0157} \approx 44.1$ days.

5. Key Report Guidelines & Common Traps

1. **Do Not Fit One Global Line:** A single global exponential fit fails because disease dynamics changed due to public health interventions, resulting in distinct growth, plateau, and surge intervals.
2. **Clear Boundary Identification:** Identify inflection dates where the slope of $\ln(y)$ visibly shifts, partition the data table into those ranges, and compute separate linear fits for each partition.
3. **Convert Equations Back:** Always convert the spreadsheet linear equations ($\ln(y) = mt + b$) back into physical exponential functions ($y = Ae^{kt}$ or $y = Ab^t$) in your report write-up.